Business intelligence and the leverage of information in healthcare organizations from a managerial perspective: a systematic literature review and research agenda

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Abstract

Purpose – Business intelligence (BI) systems and tools are deemed to be a transformative source with the potential to contribute to reshaping the way different healthcare organizations' (HCOs) services are offered and managed. However, this emerging field of research still appears underdeveloped and fragmented. Hence, this paper aims to reconciling, analyzing and synthesizing different strands of managerial-oriented literature on BI in HCOs and to enhance both theoretical and applied future contributions.

Design/methodology/approach – A literature-based framework was developed to establish and guide a three-stage state-of-the-art systematic literature review (SLR). The SLR was undertaken adopting a hybrid methodology that combines a bibliometric and a content analysis.

Findings – In total, 34 peer-review articles were included. Results revealed significant heterogeneity in theoretical basis and methodological strategies. Nonetheless, the knowledge structure of this research's stream seems to be primarily composed of five clusters of interconnected topics: (1) decision-making, relevant capabilities and value creation; (2) user satisfaction and quality; (3) process management, organizational change and financial effectiveness; (4) decision-support information, dashboard and key performance indicators; and (5) performance management and organizational effectiveness.

Originality/value – To the authors' knowledge, this is the first SLR providing a business and management-related state-of-the-art on the topic. Besides, the paper offers an original framework disentangling future research directions from each emerged cluster into issues pertaining to BI implementation, utilization and impact in HCOs. The paper also discusses the need of future contributions to explore possible integrations of BI with emerging data-driven technologies (e.g. artificial intelligence) in HCOs, as the role of BI in addressing sustainability challenges.

Keywords Business intelligence, Decision making, Health management, Healthcare organization, Information, Systematic literature review

Paper type Research paper

We wish to show our appreciation to the experts and practitioners from the field who supported us in establishing the present review (i.e. in setting the perimeter of this study and in defining the conceptual boundaries). The paper has also received constructive comments from participants at SIDREA biennial conference "*Digitalization and intelligent technologies for the governance of enterprises: the contribution of business economics to the national economic system*" (Lucca, Italy, October 20, 2022) where methodological choices and major findings have been presented and discussed. This research was conducted while Edoardo Trincanato was a Ph.D. student, and he acknowledges that his doctoral fellowship (2021-2024) was financed by The MIUR Department of Excellence Project 2018-2022 (University of Ferrara, Department of Economics and Management).

CRediT authorship contribution statement: Edoardo Trincanato: Conceptualization; Investigation; Data curation; Formal analysis; Methodology; Software; Writing-Original Draft; Writing-Review and Editing; Visualization.

Emidia Vagnoni: Investigation; Validation; Review and Editing; Supervision.

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1. Introduction

There is an ever-growing consensus among researchers, practitioners and policymakers on the extreme and far-reaching importance of healthcare fully embracing an information-oriented culture. In this pursuit, the effective gathering and utilization of health data and the leverage of information are crucial, representing a cornerstone in contemporary healthcare (Yang *et al.*, 2015; Zahid *et al.*, 2021). Furthermore, as healthcare providers continue to be tasked with doing more with less, the importance of leveraging information for control purposes and making evidence-based clinical and managerial decisions becomes even more crucial (Foshay and Kuziemsky, 2014; Basile *et al.*, 2023; Cousins *et al.*, 2023). Nonetheless, healthcare organizations (HCOs) are increasingly confronted with a myriad of technical and managerial challenges as they strive to effectively manage data and information for the purposes of informed decision-making. These challenges – strongly hindering the construction of a sustainability-oriented, anticipatory, universally accessible and data-centric digital healthcare ecosystem (Zahid *et al.*, 2021) – urgently require close scrutiny, diverse perspectives and interdisciplinary ways of knowing to foster change (McKimm *et al.*, 2020; Mayo *et al.*, 2021; Lepore *et al.*, 2023).

Among the several goals that HCOs strive to achieve is the most efficient deployment of resources while ensuring high-quality patient care (e.g. Gastaldi and Corso, 2012). Compared to other sectors, managing information in HCOs entails to deal with connected issues, such as the limited interoperability of the information ecosystem, undocked data to decision-making processes, trust and privacy issues, an overwhelming volume of data, a variety and not unified types of reporting (i.e. clinical and accounting) and a multiplicity of actors with very specific needs (e.g. Mantzana *et al.*, 2007; Mettler and Vimarlund, 2009; Brooks *et al.*, 2015; Bai *et al.*, 2018; Zahid *et al.*, 2021; Cripps and Scarbrough, 2022).

A greater employment of digital technologies in HCOs is however perceived as a viable solution to address the mentioned challenges (Gjellebæk *et al.*, 2020; Balta *et al.*, 2021; Cavicchi *et al.*, 2022; Schiavone *et al.*, 2022; Lepore *et al.*, 2023). For example, the advancement of healthcare information systems, utilized by health providers, managers and decision-makers, is increasingly considered crucial for enhancing the overall effectiveness, efficiency and reliability of services provided by HCOs (Yang *et al.*, 2015; Bai *et al.*, 2018; Ostern *et al.*, 2021). Nowadays, healthcare data can be effortlessly accessed in electronic medical records (EMR), health information exchanges and clinical data repositories. In such a scenario, information, knowledge and management control can foster decision-making, underlining the crucial role of technologies facilitating these processes (Brooks *et al.*, 2015; Laurenza *et al.*, 2018). However, as observed by Ibrahim *et al.* (2022) in a recently published systematic review of review articles on digital health, despite a well-developed body of literature investigating the role of information technology (IT) or the digitalization of medical and clinical processes, there remains a considerable lack of studies exploring digital health in areas such as decision-making, information management, cost and process efficiency, equitable healthcare and patient-centered care.

Information technologies are widely employed to enhance healthcare, extending beyond system management to rehabilitation, prevention and predictive medicine. Integration of emerging IT like big data and cloud computing offer the possibility to further reshapes and improve managerial and therapeutic approaches (e.g. thanks to the proliferation of sensors amplifies data possibilities), transforming healthcare methodologies (Yang *et al.*, 2015; Ostern *et al.*, 2021; Zahid *et al.*, 2021). Nonetheless, the current technology infrastructure in digital healthcare systems seems largely inadequate to facilitate the effective and widespread integration of emerging IT, often demanding substantial resources and very specific internal capabilities (Zahid *et al.*, 2021; Lepore *et al.*, 2023). Among the key IT, business intelligence (BI) systems and tools have been slowly rooted in the sector. BI is deemed a consistent and flourishing opportunity to address mentioned challenges related to

HCO's information systems, leveraging information coming from internal and external source, creating knowledge and informing decision-making, particularly after the COVID-19 pandemic, as underscored by the escalating number of scholarly studies and practical applications in the field (Basile *et al.*, 2023). Indeed, BI helps in analyze performance, discover new opportunities and enhance decision-making across various aspects including different stakeholders, financial matters, strategic issues and both products and services (Foley and Guillemette, 2010). Gastaldi *et al.* (2018) argued that BI has the potential to disrupt the processes through which healthcare services are offered. Recently, BI systems have attracted increasing attention from executives and decision-makers. Following the increased practice's demand for data-driven decision-making, especially from the managerial perspective (Ratia, 2018), research on BI has grown considerably over the last two decades (Ain *et al.*, 2019).

Despite the potential of BI and the growing interest, managerial-oriented research on BI in HCOs still appears to be an under-investigated and unsystematized field (Arefin *et al.*, 2020; Azevedo *et al.*, 2022; Basile *et al.*, 2023).

The aim of this paper is analyzing and synthetizing different strands of managerial-oriented literature on BI in HCOs to reveal the knowledge structure of the field, finding main and common paths to enhance both theoretical and applied future contributions by scholars from different domains. Thus, the paper is meant to providing practitioners and executives with useful insights to understand factors (barriers and drivers) critical to a successful implementation, utilization and impact of BI in different HCOs' environments.

To achieve the research aim, the article endeavors to incorporate insights arising from several studies in view of the following intermediate and connected steps: (1) to identify the research methods, theories and models that have been used by scholars (this would allow to assess the quality of the included studies and to provide methodological directions for future research); (2) to identify what the main themes of research are and to discuss the main findings (this would allow to more effectively organize and then discuss the findings from previous research, e.g. barriers and drivers of implementation and to support future research); (3) to identify gaps in the literature and to provide potential future research directions (i.e. disentangling the same from each emerged cluster into issues pertaining to BI implementation, utilization and impact in HCOs).

The paper is organized as follows: Section 2 provides a literature review on what BI is and a theoretical framework to understand BI in the context of HCOs; Section 3 describes the methodology; Section 4 presents and discusses the findings; Section 5 identifies gaps in the literature and links the findings to an actionable research agenda; and finally, Section 6 provides the concluding remarks.

2. Toward a literature-based framework to understand BI in HCOs

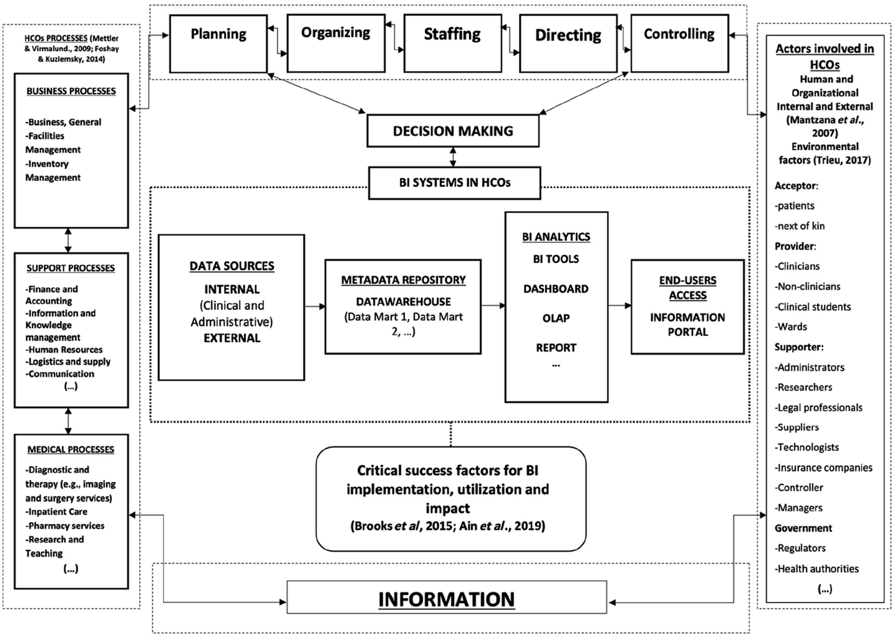
2.1 BI definition and tools

Although the term BI originated in 1989 (Negash and Gray, 2008), it is only recently that scholars and practitioners have become more deeply involved in exploring the concept (Shollo and Kautz, 2010). However, innovations and the multiplicity of technologies that over the years have been related to BI systems have led to a variety of conceptualizations and definitions of BI, which have created confusion among stakeholders and the scientific community (Foley and Guillemette, 2010). Oddly enough, rather than having a well-accepted and specific definition, BI is typically used as an "umbrella term" to describe a process, a product, a set of technologies or a combination of these (Trieu, 2017). For these reasons, given the boundaries of this research and to synthesize the findings of the articles reviewed, which emphasize different but compatible aspects of BI, we embrace the integrative and literature-based definition proposed by Foley and Guillemette (2010, p. 4):

> Business intelligence (BI) is a combination of processes, policies, culture, and technologies for gathering, manipulating, storing, and analyzing data collected from internal and external sources, in order to communicate information, create knowledge, and inform decision-making. BI helps report business performance, uncover new business opportunities, and make better business decisions regarding competitors, suppliers, customers, financial issues, strategic issues, products and services.

This definition provides a comprehensive overview of BI, making it particularly apt for informing the present review. It facilitates a more thorough understanding of the concept and aids also in leveraging BI at the strategic level in the context of HCOs. Accordingly, we focus on a wide and consolidated combination of tools that typically make up a BI system: a data warehouse (DW), online analytical processing (OLAP) and dashboards (e.g. Kimball and Ross, 2016; Ain *et al.*, 2019). A depiction of this combination representing BI is presented in the center of Figure 1, delineating four primary and interconnected blocks: data sources (both internal and external), a metadata repository (comprising the data warehouse, data mart 1, data mart 2, . . .), BI analytics, encompassing BI tools, dashboards, OLAP and reports, and, in the final "block," end-users' access, i.e. the information portal.

A DW serves as the core of a BI, consolidating information from diverse sources into a multidimensional schema to facilitate enhanced data access for analysis and decision-making (Abai *et al.*, 2013). OLAP supports real-time multidimensional analysis (Clark *et al.*, 2007), involving the presentation and querying of textual and numerical data (e.g. budgeted costs, revenue and profit) from the DW and/or other sources for analytical purposes (Jukic and Malliaris, 2008). Dashboards play a key role in visualizing data promptly, typically on a daily basis, through graphs, charts, widgets, reports and various configurations. They stand out as one of the most widely utilized performance management tools within organizations



Note(s): BI, business intelligence; HCOs, healthcare organizations; OLAP, online analytical processing

Source(s): Authors' elaboration

Figure 1.
A framework to establish and guide the present research

(Buttigieg *et al.*, 2017). Performance evaluation models and analysis tools form the basis for dashboards, which are becoming standard in hospitals (Bréant *et al.*, 2020). Generally, they are categorized into linked strategic, operational and tactical dashboards (e.g. Rasmussen *et al.*, 2009; Person, 2013; Karami *et al.*, 2013). Observing a renewed interest in dashboards nearly 40 years after their emergence, Arnaboldi *et al.* (2020) attribute this resurgence to the widespread adoption of "self-service" BI, such as desktop software like Power BI, QlikView® and Tableau™, enabling non-specialist end-users to construct dashboards on their personal computers. Dashboards include key performance indicators (KPIs), often segmented into clinical, operational and financial KPIs within the context of HCOs (e.g. Karami *et al.*, 2013).

Broadly speaking, the interoperability of BI and the mentioned tools in HCOs over the years have enabled data to be delivered to managers, administrative units and clinical staff; the interactive and user-friendly interfaces and a faster and more consistent response time have enabled users to make more use of the data (Gaardboe *et al.*, 2017).

2.2 BI in HCOs

Given the definitional premises outlined earlier regarding BI, this section complements and expands upon the previous one by delving into the specific context of HCOs in which BI is embedded. Indeed, considering BI as a combination of processes, policies, culture and technologies aimed at collecting, manipulating, storing and analyzing data from both internal and external sources, it becomes evident that its implementation, utilization and impact in HCOs are distinctly influenced the broader socio-technical information infrastructure within these organizations. This involves considering a grasp of distinct processes in HCOs, including administrative and medical procedures, an understanding of the management functions integral to HCOs' administration that influence decision-making, recognition of the diverse actors involved, and an awareness of the pivotal role played by information. Collectively, these diverse elements are therefore necessary for setting the perimeter of the present review and to disentangle and comprehend issues pertaining to BI implementation, utilization and impact within the particular context of HCOs (Figure 1). Despite the extensive body of multi-sectoral research on BI over the past two decades, encompassing numerous studies in specific areas such as adoption, utilization and success at both organizational and individual levels, there appears to be very limited exploration of all three categories (adoption, utilization and success) within a single framework. Notably, only one article (Ain *et al.*, 2019) explicitly addresses these three facets in the context of BI systems. Recognizing the value of this framework, especially in the particular context of healthcare, that is high-regulated, hierarchical, complex and multi-stakeholders, we choose to adopt a similar approach (Figure 1).

To identify the external and internal actors involved in HCOs we considered the contribution of Mantzana *et al.* (2007), because it comprehensively identified the actors involved in the adoption of information systems in HCOs (thus beyond just the users of the BI system). These actors are categorized into "acceptor" (i.e. patients, next of kin), "providers" (i.e. clinicians and non-clinicians) and "supporter" (i.e. Administrators, researchers, legal professionals, etc.). The contribution of Trieu (2017) helped to harmoniously integrate the categorization by including an environmental perspective on government and regulation influencing main actors.

To determine the BI-related processes that characterize HCOs (i.e. typically business, supportive and medical processes), we considered the influential contributions of Mettler and Vimarlund (2009), Foshay and Kuziemsky (2014) and Brooks *et al.* (2015). More specifically, these contributions suggested the first overarching frameworks to understand BI in HCOs by translating and adapting the findings of previous multisectoral research.

To identify the management functions integral to HCOs' administration that influence decision-making we include classical management functions (Figure 1). This choice deserves more clarifications. Carroll and Gillen (1987, 2019) observed that classical functions still represent the most useful way to conceptualize a manager's job. The classical functions, indeed, provide "*clear and discrete methods of classifying the thousands of different activities that managers carry out and the techniques they use in terms of the functions they perform for the achievement of organizational goals*" (Carroll and Gillen, 1987, p. 48). Particularly, Longest and Darr (2014) discussed classical management functions in the specific context of health services organizations and systems to highlight their link with the enduring relevance of decision-making in such organizations. Further, the authors outline management functions as connected with HCOs' processes and actors, which along with information constitute the blurred boundaries inside which BI is framed, interpreted and interlinked, as represented in Figure 1 blurred boundaries within which BI is conceptualized, interpreted and interconnected (Figure 1).

While the framework we have developed is tailored to the current review and its purposes, its generality allows for its application by other researchers within the HCOs context.

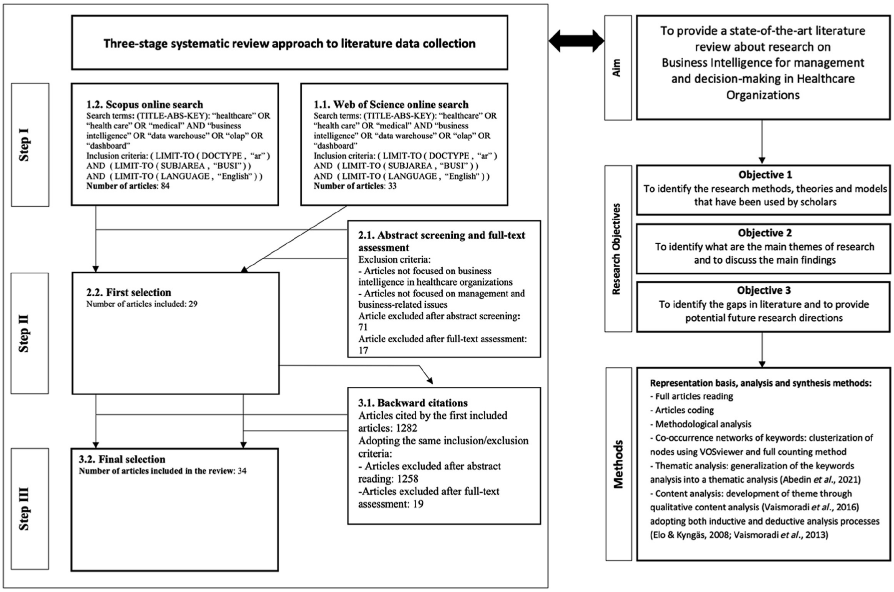
3. Research methodology

To investigate the current state of academic literature, a three-stage systematic literature review (SLR) (planning the review, conducting the review and reporting and dissemination) was undertaken in the second semester of 2022 as informed by Tranfield *et al.* (2003). The adopted approach had major steps that reflected the desirable and largely established methodological characteristics that should guide a SLR in management and social science domains, as observed by other authors (e.g. Petticrew and Roberts, 2006; Popay *et al.*, 2006). The steps are identification of the need for a review, development of a review protocol, identification and refinement of research objectives, evaluation of the inclusion/exclusion criteria, a literature search, assessment of the studies' quality and selection, data extraction, synthesis of the findings and, finally, reporting and recommendations. To limit the risk of researcher bias, a panel of experts from the authors' network was consulted to set the boundaries of the field of research, to select the keywords and to establish a set of inclusion criteria (e.g., Caputo *et al.*, 2021). Our engagement included discussions with three practitioners actively working in the field of BI within large public organizations. Additionally, we sought confirmation for the outlined framework (Figure 1) and insights from key personnel at a prominent public university hospital in Italy. This included consultation with the Director of Management Control, a manager responsible for BI within the Management Control Division, and the head of health information systems in the Pharmaceutical Department of the same hospital. Furthermore, we received valuable feedback from scholars during a national conference on digitalization and intelligent technologies for the governance of enterprises.

3.1 Search protocol and literature data

The searched databases were Scopus and Web of Science as these allow at avoiding potential biases and/or omissions and allow for future replication of our study (e.g. Caputo *et al.*, 2021). Figure 2 delineates all the steps involved in selecting publications and illustrates the methodology employed to conduct the review.

In the *first step* (step I) of the literature search, a search query was entered in the databases combining the identified search terms "healthcare" OR "health care" OR "medical" AND "business intelligence" OR "data warehouse" OR "olap" OR "dashboard." After discussion, it had been decided to identify articles that focused on "business intelligence" by taking



Source(s): Authors' elaboration

Figure 2. Steps in selecting the publications and review methodology

advantage of the keywords used by Ain *et al.* (2019) in their SLR of the last two decades of research on BI system adoption, utilization and success. Following an established approach to ensure searched publications' quality, we included only peer-reviewed articles written in English language in the field of "business, management or accounting" appearing on the databases on July 20, 2022. Further checks in both databases were done on September 16 and on December 2, 2022, and no new published articles emerged. Considering that one of the major challenges in conducting a SLR for management research is the difficulty in assessing the quality of the studies (Tranfield *et al.*, 2003), from the very beginning only peer-reviewed articles published in academic journals were included. Moreover, since BI research encompasses very different fields, the limitation to "business, management or accounting" macro area seemed the best way to reduce the high number of resulting articles and contribute to limiting possible biases in selecting only papers focused on management-related issues. Following this approach, the search returned 84 documents from Scopus and 33 documents from Web of Science.

In the *second step* (step II), the article abstracts of the identified 117 documents were scrutinized to exclude duplicates and documents clearly not focused on BI in HCOs. We obtained 46 documents that were read in full. During the full read, we confirmed exclusion for (1) every article not focused on BI in HCOs and for (2) every article not focused on management and business-related issues. Thus, all articles that discussed only the technical aspects of BI system implementation and use (e.g. in the fields of computer science or engineering) were excluded, which led 29 included articles.

In the *third step* (step III), we considered the reference lists of the included articles to collect other publications. Among the 1,282 documents obtained, after abstract screening we selected 24 articles for a full read. This procedure yielded five new articles. The result was a final sample of 34 articles. The number of included articles (34) appears comparatively analogous to recently published reviews and top tier SLR on a similar topic (e.g. Kruse *et al.*, 2017; Moore *et al.*, 2020). The authors then discussed the presence of the inclusion or exclusion criteria for every article and no significant inconsistencies emerged.

3.2 Data extraction and analysis/synthesis

To reduce human errors and biases we employed Excel software and computer-based data-extraction forms to create a data repository for the analysis. The first author read and coded all the articles (the complete list of codes is provided in Appendix). To check for reliability and consistency, the second author read and coded randomly selected articles. The two authors discussed coding in an iterative exercise, documenting all the steps taken and, again, no significant inconsistencies emerged.

The methodological section of the included studies and their quality were evaluated according to our first research objective. Complementarily, to allow an effective discussion of the articles' contributions and generalizability of their findings, we drew from Scandura and Williams' framework, that in turn integrated McGrath's (1981) dilemmatic view of the research process to comprehensively classify research strategies and their tradeoffs, namely (1) generalizability to the target population (external validity); (2) degree of precision of measurement (internal and construct validity); and finally (3) degree of realism of context.

Then, we comparatively considered results from different analyses before undertaking the second and third research steps. The process of comparison also allowed us to contrast and integrate the results, i.e. "to integrate concepts across domains into a holistic treatment of a subject" (Watson and Webster, 2020). Firstly, the data repository was scanned multiple times to describe clusters of the recurrent topics covered by the included articles, along with the results that emerged from keyword co-occurrence analysis of the "title" and "abstract" fields. In such a way, we generalized the keyword analysis into a thematic analysis following the suggestions of Abedin *et al.* (2021). Keyword co-occurrence analysis is a network-based method and a form of content analysis (Caputo *et al.*, 2018) focused on gaining understanding of the knowledge components and knowledge structure of a scientific field by examining the links between keywords. Moreover, as observed by Pizzi *et al.* (2020), such an analysis does not suffer from intrinsic bias toward older studies, which allows recent contributions to emerge. The tool that we used for the calculation in accordance with the best practices in top management journals (e.g. Raghuram *et al.*, 2019; Ali *et al.*, 2022) was the software program VOSviewer (Van Eck and Waltman, 2010). As outlined in Section 4.2, only keywords that occurred at least five times (Pizzi *et al.*, 2020) were retained, employing the VOSviewer full counting method and association-based normalization algorithm. The keywords that met the selected threshold were subsequently mapped, clustered and presented through a network diagram (Figure 5). Following this approach, we undertook the second research step. In addition, qualitative content analysis (Vaismoradi *et al.*, 2013), adopting both inductive and deductive analysis approaches over the three steps (i.e. preparation, organizing and reporting) discussed by Elo and Kyngäs (2008), was deployed. The output from each codified article was also used to understand future research directions within each cluster. These future research directions were then presented in a framework organizing the findings into issues related to BI implementation, utilization and impact in HCOs. Considering the aim of the present review, qualitative content analysis was deemed particularly suitable for the analysis and synthesis purposes. Moreover, following the methodological stance of Vaismoradi *et al.* (2016), qualitative content analysis was also used for theme development, which complemented the findings overall.

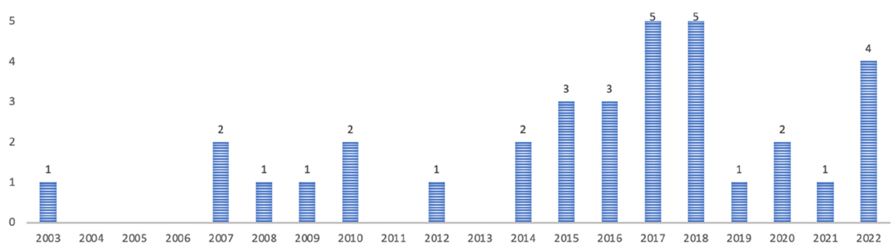
4. Findings of the systematic literature review

4.1 Descriptive results

The chronological analysis (Figures 3 and 4) aimed at displaying the spread of articles over time and the typology of academic journals. The oldest article included in the review was published in 2003, and there were only seven publications in the years 2003–2011 that, according to the data analysis, appeared more focused on BI systems' components rather than on BI systems. The increasing trend in publication clearly suggests that the topic has become more investigated recently: 70.5% (24 out of 34) of the articles were published since 2015, and there was also a notable change in the focus from the single components of BI to BI systems. The overview revealed that 5 journals contribute the most (i.e. *Decision Support System*, *International Journal of Information Systems*, *Journal of Decision System*, *Journal of Health Organization and Management* and *Journal of Knowledge Management*) with 32.4% of articles (11 articles). Overall, the presence of 5 articles (14.7%) from 4 different knowledge management (KM) journals and 10 articles (29.4%) from 5 different decision sciences and information management journals shows the relevance of the topic in these interrelated fields. Nonetheless, 67.6% of the journals (23 journals) contributed only one paper, which prove the interdisciplinary relevance of BI research in HCOs, despite the delimited perimeter of business, management and accounting macro research area, and suggests the presence of several theoretical and methodological approaches adopted by the authors.

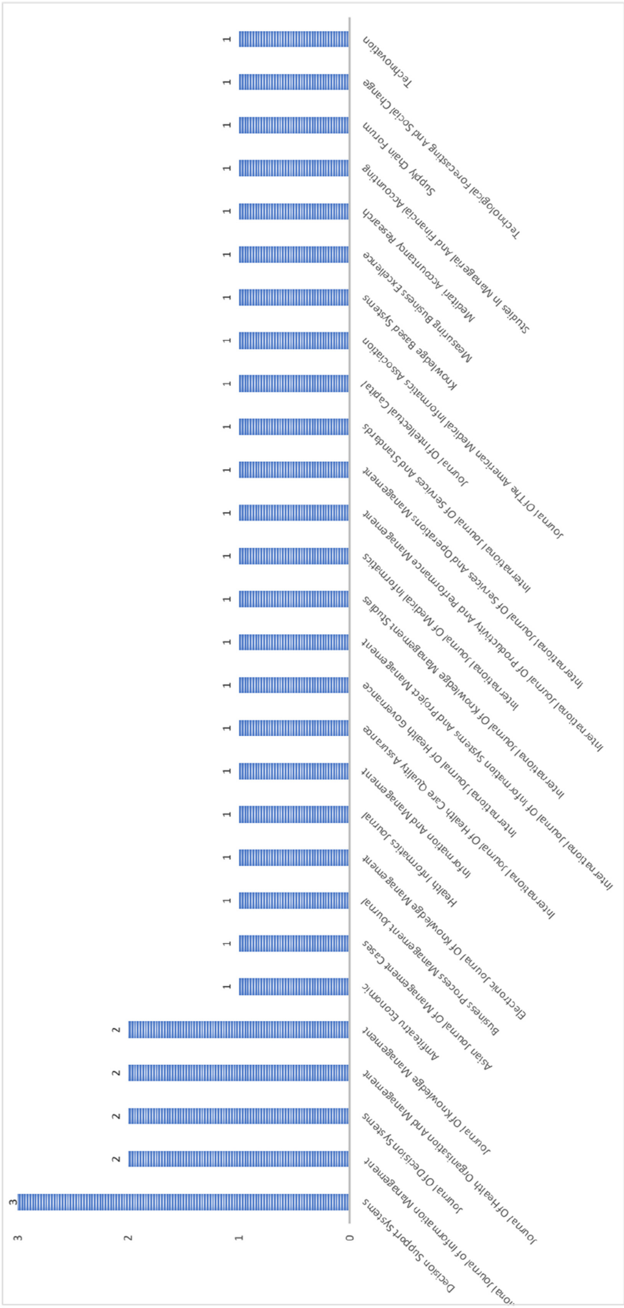
4.1.1 Methodological analysis. A methodological analysis can be very supportive in the context of a SLR because it not only allows an assessment of the studies, including critical discussion and a synthesis of their findings (Scandura and Williams, 2000), but also the identification of the most frequently used methods; this helps to provide methodological directions for future research (Saunders *et al.*, 2007). Accordingly, in this subsection we discuss the findings of the analysis carried out for each paper. Drawing on Orlikowski and Baroundi's (1991) categorization scheme for studying IT in organizations, the included articles were divided into two broad categories: *conceptual* (studies that formulate or propose concepts, models or frameworks and literature reviews) or *empirical* (studies that include forms of empirical data collection and analysis like surveys, interviews, case studies, multimethod and experiments). A strong dominance of empirical research was observed: 10 articles (29.4%) adopted a conceptual approach, and 24 articles adopted an empirical approach (70.6%), with 15 case study articles (44.1%) which makes the case study approach the most used one. As argued by several scholars who discussed the limitations of research (e.g. Mettler and Vimarlund, 2009; Olszak and Batko, 2012; Murdoch and Detsky, 2013), this represents a positive development: empirical research findings are required to understand how such organizations obtain value from BI systems.

Elaborating in this line, only 10 articles (29.4%) adopted a quantitative approach, which made a qualitative approach the most used one. Additionally, it was found that no articles



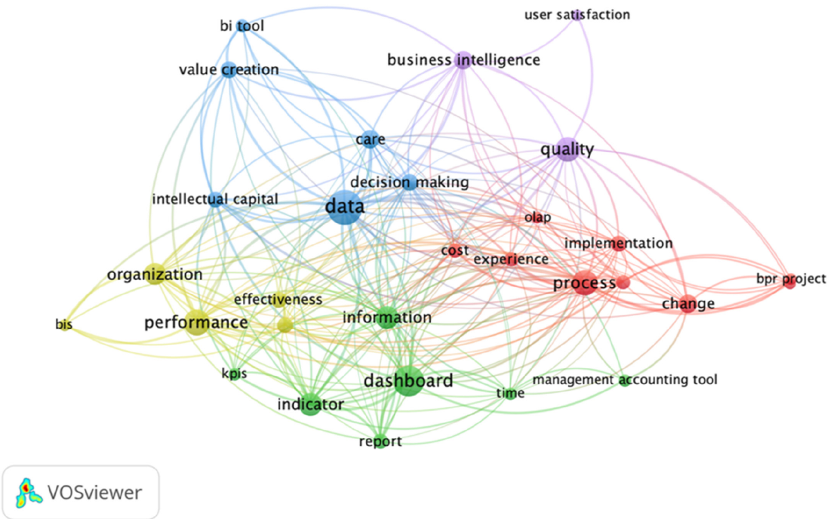
Source(s): Authors' elaboration

Figure 3.
Publications by year



Source(s): Authors' elaboration

Figure 4.
Publication by outlet



Source(s): Authors' elaboration

Figure 5. Network diagram of the co-occurrence of keywords from "title" and "abstract" fields

used a mixed methods methodology and that only three articles (8.8%) adopted an interpretative approach.

Regarding the generalizability, the underlined dominance of empirical research and case studies approaches calls for close understanding of other variables – like the research setting or the typology of organizations, the level of the analysis and the types of data analyzed by the authors – to allow an effective discussion of the articles' contributions and to synthesize the findings. As outlined in the previous section, we collected these data drawing from the contribution of Scandura and Williams (2000), which in turn integrated McGrath's (1981) dilemmatic view of the research process, to comprehensively classify research strategies.

4.2 Five clusters and main findings of management and decision-making related research on BI in HCOs

As presented in Section 3, a bibliometric analysis was conducted and five clusters identifying the main topic or research themes emerged. Only keywords that occurred at least five times were kept using the VOSviewer full counting method and association-based normalization algorithm. The keywords that satisfied the selected threshold were then mapped, clustered and presented through a diagram of the network (see Figure 5). As done by Radhakrishnan *et al.* (2017), data were cleaned before the creation of the diagram network to avoid redundancy. Indeed, considering that every included article is about BI in HCOs, all the keywords related to "HCOs" like "healthcare," "health," "hospital" or "medical" were excluded since they obtained the highest ranking and were consequently expected to affect the formation of clusters.

4.2.1 Cluster 1 (blue): data-driven decision-making, relevant capabilities and value creation. According to the integrative findings of the deployed content analysis, this cluster represents the widest and most interconnected areas of research. This comes as no surprise considering that BI can be seen as always oriented to create knowledge and to support some decision domains.

The most dated article included in this cluster was the contribution by Berndt *et al.* (2003). As argued by the authors, among the pivotal "missions" of a DW are the support of decision-making activities and the modeling of an infrastructure for ad hoc exploration of very large collections of data without the intermediation of database programmers. Accordingly, the emphasis is on the understandability and intuitiveness of data navigation for decision-makers and other end-users. Nonetheless, as observed by Berndt *et al.* (2003), the capabilities required to successfully implement BI systems in HCOs, and take advantage of it, were not forgotten. Along these lines, Berndt *et al.* (2003) predicted the relevance of BI systems and tools to open new research avenues about decision-making and value creation in healthcare. Over the years, many issues have arisen at the intersection of BI, such as relevant capabilities of end-users, healthcare decision-making and value creation. More specifically, we focus on those insights from the literature that discussed interrelations, even in the context of group decision-making, ramifications of political processes in relation to decision-making, or dissemination patterns of healthcare information in the community.

Regarding DW and health networks, von Lubitz and Patricelli's (2007) article was among the first to emphasize that a federated approach to data storage and management is key to improve access to coherent, relevant and timely information allowing "*better, more cost effective and medically sound access, delivery and administration of global healthcare*" (von Lubitz and Patricelli, 2007, p. 97).

Tremblay *et al.* (2007) focused their attention on another BI tool: an OLAP interface on the CATCH DW used by knowledge workers at a regional health planning agency in Florida. Tremblay *et al.* (2007) found that OLAP can make strong contributions to decision-making and performance, especially in the data-intensive environment of HCOs. Looking at the big picture (i.e. observing that at that time the existing research about BI was primarily focused on the industrial sector), Mettler and Vimarlund (2009) provided the first overarching framework to understand BI in HCOs by translating and adapting the findings of previous multisectoral research. Similarly, Brooks *et al.* (2015) provided the first framework for developing a domain-specific BI maturity model in the context of healthcare, to address highly important information needs to understand industry-specific requirements and complexities.

With a different approach, Safwan *et al.* (2016) discussed one of the first case studies on BI development in a healthcare setting and highlighted why BI can be considered a real boost to improve decisions in HCOs, especially the decisions made by physicians. Safwan *et al.* (2016) showed the evolutionary nature of BI systems and emphasized two factors that can limit the development of a BI project in such contexts: (1) the fact that its nature does not allow it to be initially developed as a large-scale system and (2) the emergence of several challenges regarding the support required by both clinical decision-making and hospital management. Safwan *et al.* (2016) observed that "*the difference in focus between operational clinical decision support, versus the financial and strategic focus of hospital management decision support, was too distinct for the one team to manage*" (p. 17). Similarly, Wang and Bird's (2017) results showed that data interpretation is a crucial capability that directly influences decision-making effectiveness. Discussing the importance of relevant professional skills for BI, Wang *et al.* (2018) called for attention to the way in which big data analytics capabilities transform organizational practices in HCOs and generate value and potential benefits. Through the lens of IC, Ratia (2018) discussed to what extent internal and external data utilization and requires specific capabilities, such as IC, as different data sources (structural capital) and organizational competences (human capital) can have an impact on the organization's value creation. However, Ratia (2018) concluded that existing organizational capabilities about BI utilization in observed cases were insufficient to fulfill organizational ambitions. Ratia (2018) pointed out the constant need to seek external capabilities in BI utilization within the analyzed Finnish private HCOs; hence, internal capabilities should be fostered or created to enhance human capital and enable efficient future data processing. This step forward requires changing the mindset on the use of data and required ad hoc capabilities (Ratia *et al.*, 2019).

From this perspective, Basile *et al.* (2023) provided insightful hints about ethical-related issues on BI, relevant capabilities and decision-making. Theoretically, Basile *et al.* (2023) proved that "*the exploitation of data through BI in the decision-making process can outperform experience-driven practices for managing processes in the healthcare domain*" (p. 2). However, as observed by the authors, data-driven technologies are different from artificial intelligence (AI) technologies, since the former are used to improve the cognitive and calculation capabilities of humans, not to mimic their capabilities. From a managerial perspective, the model implemented by Basile *et al.* (2023) demonstrates that BI can help physicians to make best decisions, and improve the effectiveness of the decision-making process, leading also to financial savings.

4.2.2 Cluster 2 (purple): user satisfaction and quality. According to the World Health Organization (WHO, 2010, p. 44), information system in HCOs have four key functions: (1) data generation; (2) compilation; (3) analysis and synthesis; and (4) communication and use. Health IS collects data from several sources, analyzes the data and ensures their overall quality, relevance and timeliness to support decision-makers. As highlighted by Bai *et al.* (2018), data quality and interpretation issues in HCOs impact all these functions, as well as the final quality of the services provided (Martin *et al.*, 2022). Moreover, issues centering on the satisfaction of BI end-users and patients are deemed as pivotal in the literature and, from this perspective, hospital dashboards can play a crucial role (Buttigieg *et al.*, 2017; Pace and Buttigieg, 2017).

Papers included in the present cluster discussed issues pertaining to (1) the *satisfaction* of end-users with BI system in HCOs and its implications or, in parallel, issues pertaining to the satisfaction of patients; (2) the *quality* of data or the quality of BI systems and tools, and its implications for the quality of the services provided.

According to our articles' sample, the most dated contribution in this area was that by Tremblay *et al.* (2012) who discussed the management of outliers or data errors presented by reporting tools (e.g. OLAP) in HCOs. By considering the different types of data in HCOs (i.e. clinical and accounting data), the focus of Tremblay *et al.* (2012) was on providing useful insights to improve HCOs' practices, patient outcomes, quality of care and to complement BI tools to better support users' understanding and satisfaction. Users' satisfaction is crucial for information systems' success. For example, Gaardboe *et al.* (2017) used DeLone and McLean's IS success model as a framework to analyze data from 1,352 respondents among BI users in public HCOs in Denmark; they found that system quality was positively and significantly associated with both use and user satisfaction. Moreover, information quality appeared positively and significantly associated with user satisfaction and, finally, user satisfaction also appeared positively and significantly related to individual impact. According to this stream of literature, the effectiveness, efficiency and reliability of HCOs' services is strongly influenced by the quality of data in the health IS used by HCOs, managers and decision-makers (Bai *et al.*, 2018).

4.2.3 Cluster 3 (red): process management, organizational change and financial effectiveness. Nowadays, HCOs face the growing need to develop effective management and control systems due to both internal and external pressures (Ippolito *et al.*, 2022). Several articles discussed issues pertaining to the assessment or the reduction of costs through BI systems' adoption and BI tools (e.g. Basile *et al.*, 2023; Ratia *et al.*, 2018; Bittencourt *et al.*, 2017; Muriana *et al.*, 2016). However, articles in this cluster also had a clear focus on the processes that characterize HCOs (business, support and medical) and their strategic control to foster organizational change and financial effectiveness.

Ferranti *et al.*'s (2010) article is one of the early contributions in the context of leveraging BI tools in support of patient safety and financial safety; they underlined that through BI and analytics, HCOs can transform unwieldy amalgamations of data into information that can help to improve patient outcomes, increase financial safety and enhance operational efficiency. Interestingly, Ferranti *et al.* (2010) also stressed that BI systems and tools help users to understand and to become aware of how their actions affect the entire organization. With a similar focus on organizational processes, Dwivedi *et al.* (2008) proffered a knowledge management (KM) solution, contending that an integration of KM and ICT into HCOs' specific organizational processes will lead to more effective and efficient operations.

Along these lines, Ghannouchi *et al.* (2010) observed that introducing and fostering organizational changes in HCOs' processes is increasingly perceived as a necessity to offer faster services and better quality. Through a case study approach, the authors discussed the contribution that DW technology can bring to business process reengineering projects and how it can improve their success. They also discussed the peculiarities of such projects using DW technology and demonstrated the great importance of a DW introduction for the improvement of HCOs' processes. In the outlined context, Ghannouchi *et al.* (2010) finally reported a favorable vision of senior management since an increased decentralization of decision-making power and centralization of data storage/archiving allowed a greater and more effective control and diagnosis of an HCO's processes.

Moreover, from a strategic control perspective, hospital capacity planning is a key subject in the management of HCOs as it deals with the problem of scarce resources and their most efficient allocation and use. In this regard, Yip *et al.* (2018) presented a simulation model aimed to improve patient flow, capacity management and resource allocation through leveling bed occupancy within the hospital. Similarly, on the assumption that (1) in the short term the main challenge for hospitals is to employ their facility efficiently, (2) control of the patients flow occupying hospital beds is crucial to ensure sustainability, Bittencourt *et al.* (2017) recommended a dashboard to monitor operations (daily) related to admissions, capacity and discharges.

It is finally stressed that one of the key elements in developing an established culture of change management and accountability in HCOs is monitoring KPIs (e.g. Ippolito *et al.*, 2022). However, as observed by Bergeron (2017, p. 2), "*there is more to performance management than selecting a few KPIs from a list and feeding them into a graphical dashboard system. It's about behavior change, leadership, and vision*."

4.2.4 Cluster 4 (green): decision-support information, dashboard and KPIs. The present cluster is characterized by a stream of literature providing insights about dashboards' implementation issues and challenges in different HCO environments. More specifically, Kumar *et al.* (2020, p. 12) observed that since healthcare IT resistance "*is embedded in the interaction of people, practice, and technology*," it is essential to examine such resistance by engaging with the distinct issues of this interaction in each specific context. Adopting this kind of perspective appears pivotal to leverage BI in HCOs and hence to enhance their efficiency in the information management for decision-making (Brooks *et al.*, 2015).

A contribution amenable to this cluster is the one of Foshay and Kuziemsky (2014, p. 26) since it provided a theory-driven framework "*for information needs identification to support healthcare processes*." The framework identified core HCO processes (medical, business and support) and seemed to be able to guide the definition and prioritization of decision-support information needs in such contexts. The contribution of Foshay and Kuziemsky (2014) also dealt with issues pertaining to implementation factors, the lack of decision support capabilities, the quality of operational data and related processes, and the individual-oriented root causes.

Dowding *et al.* (2015) aimed to lay out a comprehensive overview of the state of evidence for the use of clinical or quality dashboards in HCOs from 1996 to 2012. The authors found some evidence that implementing clinical and/or quality dashboards can improve adherence to quality guidelines and may help to improve patient outcomes, but it is unclear what dashboard characteristics are related to these results. On this line, Buttigieg *et al.*'s (2017) comprehensive analysis of hospital performance dashboards for the period 2000–2016 found that achieving an effective dashboard in any setting required significant efforts and significant investment in financial and human resources. However, despite the fragmented evidence provided by the literature, along with several challenges, Buttigieg *et al.* (2017) listed many advantages generated by performance dashboards in HCOs (e.g. automation in data collection and transference). Against this backdrop, different articles included in this review substantially confirmed the above outlined findings (Rana, 2015; Nagel-Piciorus *et al.*, 2016; Bittencourt *et al.*, 2017; Pace and Buttigieg, 2017; Schiavone *et al.*, 2022).

More in depth, Nagel-Piciorus *et al.* (2016) presented the experience of an HCO on the development of an integrated management and reporting system, which meant to merge distinctly regulated management systems (e.g. for quality, risk, IT, health or environmental management) with strategic planning and control via a balanced dashboard and BI tools. Emphasizing the importance of international standards to increase the quality of services (e.g. ones from the ISO 9000 family), Nagel-Piciorus *et al.* (2016) discussed the advantages from the merging of "usually unbridged" management modules, as well as the importance to clearly define objectives, at the intersection between strategy, mission and business purposes.

There are several improvement facets that need to be considered for the development of a dashboard in an HCO, and it appears difficult to reproduce the findings in different contexts. However, as observed by Pace and Buttigieg (2017), who discussed hospital dashboards' visibility of information at different management levels to improve quality and performance at an acute general hospital in Malta, most of the results obtained could be matched with the current literature. Data collected by Pace and Buttigieg (2017) were organized into three main themes: (1) levels of information visible on dashboard; (2) visibility of information from board to bedside; and (3) visibility of information outside hospital. Accordingly, the study discussed the different levels of information that should be visible from bedside to board (i.e. to management, clinicians and public) and it still represents, to our knowledge, the only contribution that deals with such issues in HCOs. Complementarily, Ippolito *et al.*'s (2022) findings focused on the crucial role played by technology in performance measurement and management in HCOs, as well as on the need to actively involve middle management in such processes.

Schiavone *et al.* (2022) explored the building of a dashboard to monitor KPIs in digitized health networks in Italy. By developing an integrated "meso-framework" based on the centrality of IC components, the latter is one of the few studies that attempted at analyzing the performance and evaluation of HCOs' networks. Moreover, it was underlined that Schiavone *et al.*'s (2022) conceptual model could be useful to classify data; they observed that their model embraced a BI application as *information* needs to be interpreted using a reliable framework that results in the gathering, sharing, analysis and monitoring of information.

4.2.5 Cluster 5 (yellow): performance management and organizational effectiveness. Articles included in this cluster are focused on management control systems (MCS) in HCOs and, more specifically, on BI's impact at the organizational level and performance management. Nonetheless, rather than evaluating management accounting tools like dashboards based on their usefulness to support decision-making or control, Flachère (2014) and Arefin *et al.* (2020) discussed the organizational and social role – as mediator – that they can play in HCOs to organize new managerial practices. Flachère (2014, p. 95) observed that the use of dashboards – for both clinical and managerial activities – appeared as most often limited to observation rather than action or corrective behavior, and that top management do not use them "to pressurize" clinical units since the transfer of responsibilities did not occur. However, according to the author findings, dashboards helped chief physicians to have a better vision of their clinical units and to know where the units stand, compared to previous periods, also helping to set new management practices and dialogues. Therefore, the use of management accounting tools within the hospital and between heterogeneous actors has fostered "decompartmentalization" between medical vis-à-vis administrative logics: "*head* *physicians have moved from a purely medical vision to a more* 'medico-economic vision,' *while some top management members have moved from an* 'economic' *vision to a* 'medico-economic' *vision. In this way, head physician and top management representations have evolved towards a shared representation of clinical unit management that of a* 'medico-economic' *representation*" (Flachère, 2014, p. 96). Along similar lines, Gravili *et al.* (2020) discussed the value of IC and big data to assess performance in healthcare. They argued that "*a very significant part of knowledge is tacit, that is, scarcely codifiable, spreadable and appropriate by third parties and which makes it less tacit, but if made available and pervasive, it could produce new value*" (Gravili *et al.*, 2020, p. 24). Similarly, Arefin *et al.* (2020) assessed the effectiveness of the mediating role of BI in HCOs by more specifically examining organizational learning culture as the antecedent and organizational performance as the consequence of BI.

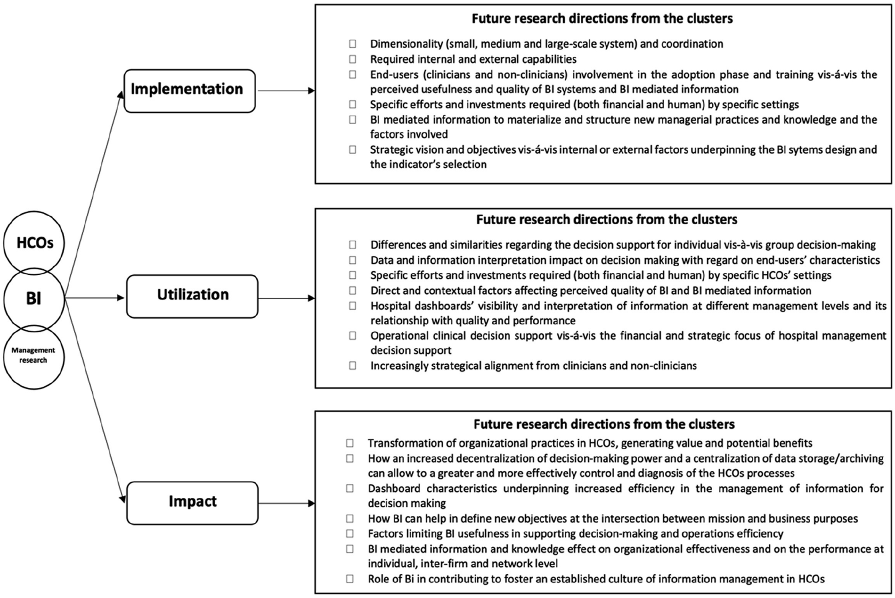
Thus, as observed by Muriana *et al.* (2016), HCOs' performance assessment involves the consideration of non-stationary KPI behavior and the holistic nature of such performances. Bréant *et al.* (2020) argued the holistic nature of performance management in HCO environments by moving from the assumption that multidimensional dashboards are required to provide to the stakeholders the tools needed to improve both clinical and accounting processes. Furthermore, the authors discussed the organizational and IT approaches that allowed, over the previous ten years, the evolution of the Geneva University Hospital's decision support environment into a "*scalable, productive and sustainable*" BI ecosystem (Bréant *et al.*, 2020, p. 1). The authors underlined how the decision support environment of such institution have been increasingly aligned with the strategic objectives, supporting the clinical and administrative staff involved in control functions by providing actionable knowledge.

5. Discussion and future research agenda

The emerged clusters of research exhibit both distinct thematic characterization and significant interconnections. As emerged from the findings, indeed, the knowledge structure of this research's stream seems to be primarily composed of five clusters of interconnected topics: (1) decision-making, relevant capabilities and value creation; (2) user satisfaction and quality; (3) process management, organizational change and financial effectiveness: (4) decision-support information, dashboard and key performance indicators; and (5) performance management and organizational effectiveness.

These clusters represent key thematic areas that form the core of research in BI, with each cluster contributing to a nuanced understanding of the adoption, utilization and impact of BI in HCOs. As the current literature review was also meant at identifying gaps in the literature and to contribute to identify an actionable research agenda to enhance scholars' contributions, we disentangle several future research directions from each cluster. These have been organized (as shown in Figure 6) in issues pertaining to BI implementation, utilization and impact, in connection with the presented framework synthesized in Figure 1, thus also considering the processes, actors and management functions involved.

As discussed in the context of the outline framework (section 2), considering BI as a combination of processes, policies, culture and technologies designed to gather, manipulate, store and analyze data from internal and external sources, it becomes apparent that its implementation, utilization and impact within HCOs are intricately shaped by the broader socio-technical information infrastructure in which BI in embedded. In this regard, cluster findings confirm that challenges associated with regulation, diverse needs and expertise of multiple stakeholders, the intricate nature of the healthcare sector, and the prevailing hierarchical decision-making structures within HCOs underscore the importance of understanding context-specific critical success factors influencing BI.



Source(s): Authors' elaboration

Figure 6. Future research directions from the cluster

In terms of implementation challenges, findings from the clusters confirms that the intricate socio-technical information infrastructure, combined with interconnected issues stemming from the participation of diverse stakeholders with unique needs and different background, strict regulations and limited interoperability, makes the task of implementing BI in HCOs, especially in public institutions, notably different from other contexts. In this regard, as summarized in Figure 6, there is the need for additional theoretical and applied contributions. This holds also true for issues related to the utilization and impact of BI in HCOs, where there is a clear need for further exploration. Specifically, in addressing critical success factors pertaining to utilization, future research is highly encouraged to enhance the experiences and strategic alignment of clinicians and personnel without a managerial background, also considering the usage, yet whether it is individual or at a team-level. Regarding impact, among the various identified directions for future research, it is crucial to delve into the diverse and interconnected ways through which BI can generate value in HCOs to encourage its adoption, particularly in light of the growing challenges these organizations are confronted with. By and large, future researchers can leverage insights from contributions in multi-sectoral literature on BI implementation, utilization or impact-related issues to comparatively discuss potential differences or applications in the specific context of HCOs. Emerging data-driven technologies present also significant opportunities for BI in HCOs, offering avenues to address sustainability objectives in these organization. Almost all of these particular subjects have been overlooked in extant healthcare research on BI.

Considering the results of the analysis carried out, future interested researchers could consider:

(1) *From a MCS' perspective*, a common thread emerged from the present research; most of the articles discussed the need of HCOs to conjointly improve both the quality of their services and their efficiency, while controlling the costs and the best allocation of resources. Considering that nowadays information in HCOs and digitalization are

increasing at a tremendous speed, originating from multiple sources at different levels, the analysis clearly reveals that HCOs also require the introduction of capabilities, technological solutions and tools in a way that complements their MCS. This would enable HCOs to better detect and analyze problems and to provide solutions that cumulatively impact the value that they create. As stated, 70.5% of the articles were published from 2015 onward with a prominent change in focus from the single components of BI to BI system, and on an increasing attempt to reach alignment with MCS and strategic objectives. From this perspective, BI systems and tools emerged as a consistent and flourishing opportunity to address discussed challenges in HCOs, especially when the potential of BI-mediated information is fostered by control systems design and therefore leveraged to support control and decision-making processes.

(2) *From an information management and KM perspective,* in all clusters different contributions emphasized that BI cannot always provide benefits to HCOs' processes due to organizational culture issues, scant strategic alignment and vision, lack of competences or management support. Despite this, several studies emphasized other ways through which BI can be beneficial for HCOs, such as the possible organizational and social role of BI within HCOs and data-intensive healthcare. An example is the production of actionable knowledge or the fact that BI-mediated information and BI tools can contribute to make knowledge more available and therefore pervasive, which contributes to the production of new value. The study by Gravili *et al.* (2020) revealed that the analysis of big data in HCOs can be a means for its formalization and structuring. Indeed, Gravili *et al.* (2020, p. 4) concluded in relation to the value of IC and big data to assess performance in healthcare, "*a very significant part of knowledge is tacit, that is, scarcely codifiable, spreadable and appropriate by third parties and which makes it less tacit, but if made available and pervasive, it could produce new value*." Along these lines, some contributions discussed the BI potential to foster connections among different departments and roles that were previously separated, impacting the dynamics of knowledge creation and diffusion in HCOs. Thus, the social role of BI is emphasized. Moreover, actionable knowledge and BI-mediated information can have a positive impact at different levels. Future research could therefore integrate further methods and models from these interconnected fields, where evident research gaps have been found at an individual, organizational and especially at a network level of analysis.

(3) *From a sustainability perspective*, none of the articles directly addressed the urgent sustainability needs in relation to BI implementation, utilization and impact in HCOs, although BI systems and tools clearly show the potential to address these challenges, as highlighted by multisectoral literature (e.g. Petrini and Pozzebon, 2009; Olszak *et al.*, 2021; Azevedo *et al.*, 2022; Seddigh *et al.*, 2022). However, despite the absence of articles explicitly addressing sustainability, the transformative potential of BI systems and tools to counteract environmental and social challenges appears evident in HCOs due to the topics covered and highlighted impacts. Therefore, the possibility offered by BI in HCOs to address sustainability challenges represents an opportunity for both theory and practice.

(4) *From the perspective of emerging data-driven technologies*, our examination reveals a notable absence of published articles integrating key components such as big data, IoT, AI, machine learning or neural networks into managerial-oriented on BI in HCOs. This gap encompasses studies on BI implementation, utilization and impact. These technologies have the potential to integrate and support BI at various levels in HCOs,

hence presenting new opportunities but also underappreciated challenges that necessitate context-specific considerations. For instance, in data extraction, as the technological landscape evolves, data mining becomes increasingly relevant. Techniques such as clustering, data summarization, learning classification rules, discovering dependency networks, analyzing changes and detecting anomalies play a crucial role in extracting implicit, previously unknown, and potentially valuable information from data. This represents a domain that warrants further exploration for managerial oriented research, particularly in healthcare, where complexity, strict regulation and scarce interoperability limit data access and hinder informed decision-making. Moreover, emerging technologies can facilitate the utilization of BI solutions. Automation of complex data processing tasks and enhancement of end-user access are potential benefits, since visualization-related challenges persist, especially for clinicians and administrative staff lacking an analytic background.

6. Conclusion

Nowadays, HCOs face several burdensome challenges in relation to which BI systems and tools are revealed to be a transformative source with the potential to contribute, despite some hindrances, to reshaping the way different HCO services are offered and managed. Our analysis allowed us to reconcile fragmented and early-stage literature. On this matter, the ratio of empirical (70.5%) to conceptual (29.4%) articles could encourage academics and practitioners to investigate this emerging field through more theoretical lenses. Empirical research findings are undoubtedly required to begin making substantial headway, but both practitioners and scholars require overarching frameworks to compare and integrate the findings (i.e. to guide future and more tailor-based applications and research).

From a theoretical standpoint, our article indeed contributes in various ways. Following a rigorous inclusion procedure to gather only peer-reviewed articles entirely focused on HCOs, critical findings were quantitatively mapped out into five clusters that allowed the main research themes to emerge. To allow such descriptive analysis to contribute theoretically (i.e. by materializing the knowledge structure of this emerging stream of research), a theory-driven qualitative content analysis was performed alongside to evaluate, aggregate, systematize and discuss the resulting clusters. The analysis of each cluster brought to identify an actionable research agenda that comprehensively discussed: (1) future research directions and, closely related, (2) a further theoretical synthesis in view of MCS, information management, KM, sustainability and emerging data-driven technologies research perspectives. Practitioners and executives can benefit from this work, which allows for raising awareness of the multifaceted contribution of BI and its implications, even when BI systems are adopted for a specific purpose in the context of HCOs. From such interpretations and in line with the increasingly advanced complex techno-social systems within the healthcare environment, they can also take advantage of the practical insights that have emerged, as organized in each cluster, i.e., in relation to critical success factors for the implementation and utilization of BI in different HCOs' environments.

Finally, the current research is not without limitations, as it brings intrinsic limitations associated with methodological choices. A first limitation concerning sample selection may be represented by the identified keywords, despite taking advantage of keywords and categorizations proposed by previously influential SLRs and expert assessments. A second limitation may be associated with the prevalent type of research approach detected, the case study and a variety of empirical studies dealing with specific contextual factors or complex dynamics. This may have limited the discussion of the findings beyond efforts in coding and analysis to integrate the challenging framework proposed by Scandura and Williams (2000).

Thirdly, although rigorously developed through both quantitative and qualitative approaches for analysis and synthesis, it is also acknowledged that clustering of the articles can always be seen as the result of subjective decisions, thereby appearing liable for further development.

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Appendix

List of codes used in the present review

Bibliographic information (e.g. author, title, date of publication, etc.)
Key focus on HCOs
Key focus on business intelligence
Key focus on management research and practice
Types of organization (e.g. public hospital, private hospital, university hospital, hospital networks, etc.)
Research setting/country and level of development
Theories/framework/model used
Methodology/strategies
Types of data

Objectives/research questions
Key findings and relevance for adoption, implementation and impact
Themes
Cluster
Level of the analysis (e.g. governance or board level, middle or top management, etc.)
Gap for future research identified by the authors
Personal notes and details of synthesis

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